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# A machine learning model with linear and quadratic regression for designing pharmaceutical supply chains with soft time windows and perishable products

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ABSTRACT

Product perishability is an important problem in Pharmaceutical Supply Chains (PSC). Demand perishability and unpredictability add enormous complexity to successfully implementing efficient Pharmaceutical Supply Chain Network Design (PSCND). This study develops a mathematical model for the PSCND. The two objective functions used in the model minimize the total cost and the delivery penalty due to schedule violations involving Soft Time Windows (STWs). Using a STW strategy in the Distribution Center (DC) allocation model empowers decision-makers to include a time-based performance metric for DC evaluation based on the degree of urgency or need for a part. Moreover, Linear Regression (LR) and Quadratic Regression (QR) Machine Learning (ML) algorithms are proposed to forecast the demand and decrease the possibility of a shortage in the PSCND. We show that QR has better performance than LR in PSCND. In the proposed approach, the demand for medicine is forecasted by the QR technique. The Goal Attainment (GA) method is used to solve the suggested model. Finally, sensitivity analysis and managerial perspectives are offered. The numerical outcomes show that the suggested model leads to an efficient PSC, reducing cost, decreasing shortage, and increasing customer satisfaction with STW consideration.

## 1. Introduction

Among healthcare products, medicines account for 20%–30% of global health expenses [1]. Pharmaceutical Firms (PFs) in the past disregarded the idea of Supply Chain Management (SCM). However, a number of challenges are forcing PFs to alter their established management strategies in order to obtain a competitive edge by concentrating on the Supply Chain (SC). The majority of studies on the subject of PSCs deal with efficacy and safety. However, controlling PSCs has not received much attention [2].

The already medical networks are under more strain as a result of COVID-19. Reducing the occurrence and effects of shortages requires improved SC [3]. Therefore, improving the PSC is essential for the healthcare sector. The PFs is engaged in drug investigation, development, manufacturing, marketing, and giving prescriptions to clients for vaccination, recovery, or symptom relief. The PSC is a way through which proper medicines are distributed at a suitable time and place to reach final customers [4]. The global healthcare expenditures boost around 9.8% of the global Gross Domestic Product (GDP) in 2019 with a $8.5 trillion valuation [5]. Therefore, disruption in production or distribution can jeopardize the lives of humans by making problems in delivering drug items [6]. Since customers are a major part of each SC, especially the PSC, satisfying their demand is indicated as one of the main goals of the SCs [7]. In this regard, one of the most important concepts in the SC is responsiveness. The SC's responsiveness is described as its capacity to meet client demand in a shorter amount of time [8,9].

The use of optimization tools in the PSCN area has aroused a penchant for researchers. Today's businesses use lean strategy in SCM to eliminate waste, enhance efficiency, and, thus their competitiveness. The five high-level organizational efforts that make up the lean strategy are: (i) plans for improving suppliers, (ii) waste reduction, (iii) ongoing improvement, (iv) flawless quality, and (v) supplier flexibility. It is evident from the literature and practices that one strategy to reduce waste is to produce and store the required amount and to use Lean Procurement (LP) [10]. To ensure that the production line receives timely access to its supply of parts, suppliers participating in the LP program concentrate on timing and order quantity. Under LP, early and late deliveries will result in rising in inventory or the shutdown of the whole assembly line, which will reduce system performance. Grout [11] has placed a special emphasis on the timely delivery element by highlighting the significance of cutting down on waste. For on-time

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delivery, a selection incentive model was put forth. We notice and infer from experience that the requirement for precise, on-time delivery for LP is unrealistic and unworkable. Therefore, some sort of mechanism is required to practically enable LP to function well under real-world circumstances. Thus, one contribution of this work is modeling the realistic circumstance whereby the DC can deliver within a specific Time Window (TW) i.e., a STW strategy is introduced for the first time in this network. This study suggests a flexible time-window technique where it is acceptable for DCs to deliver parts within a reasonable time span. Another issue that is important in PSC is the perishability of the product. A medicine's shelf life is the period of time during which its stability is greater than 90% [12]. The utilization of discarded or expired drugs at any stage along the PSC is referred to as "medicinal waste" [13]. Moreover, this waste costs healthcare organizations a big amount of dollars each year owing to, unneeded drugs, disposal expenses, and spending time on activities such as advising and prescribing [14,15]. So, considering shelf life in pharma chain is important and decreases wastes that this study have considered this issue. Moreover, for PFs, demand forecasting is essential for two reasons: first, any imbalance between demand and supply could have an adverse effect on patients and the medication distribution chain, potentially putting their lives in danger; and second, patients who cannot tolerate unpredictable order fulfillment may switch to an alternative medication, which could result in a patient's permanent lost sales if a demand is not met [16].

In the literature, different papers have written about PSC [17,18], but they were failed to achieve customer satisfaction despite advancements made in production and distribution techniques.

Therefore, this study focuses on the PSCN design to increase customers' satisfaction and responsiveness of the system while simultaneously reduce the cost and waste of the PSC, and it develops a mathematical model to solve the PSCN issue of LP with a flexible time-window strategy considering perishability executed. To predict the demand of customer two ML techniques are implemented to increase accuracy and resiliency of model, first time in this work. Finally, the GA method is suggested to solve the new Bi-Objective Integer Linear Programming (BOMILP) model. This research by considering two objective functions, several constraints, and using ML algorithm attain those objectives.

The reminder for the work is set up as follows: The summary of literature was declared in Section 2. Section 3 illustrates the ML technique for predicting demand. The problem description and its model structure are indicated in Section 4. Section 5 presents the solution method. The output of the numerical result, sensitivity analysis, and managerial perspectives are declared in Section 6. Finally, the result and future suggestions are provided in Section 7.

## 2. Literature review

Savadkoohi et al. [19] established a three-layer PSCND problem using a multi-period, multi-product location-probabilistic inventory model. The concept was created to pay for the entire network. Sieckmann et al. [20] outlined lean success elements and challenges and suggested a process for implementing LPS that takes into account the special characteristics of SMEs in the pharmaceutical sector. Particular emphasis is placed on taking human-oriented elements into account and choosing lean techniques that are in line with corporate objectives. Jabbarzadeh et al. [21] illustrated a robust optimization model that considers production and distribution planning of a SC Simultaneous within the postponement strategy. They found that the mentioned strategy can consistently save money for the SC. Merkuryeva et al. [22] addressed cutting-edge techniques and significant issues with demand forecasting for the PFs. Viegas et al. [23] recommended three categories of reverse flows: donations, reverse logistics, and circular economy. Contributions are distinguished by overt acts of philanthropy that affect a company's brand or by urgent humanitarian intervention. Tat et al. [24] analyzed a two-echelon PSC, including a single pharma-supplier and a single pharmacy with one type of fixed shelf-life medicine. A scheme for reducing pharmaceutical waste is illustrated to decrease cost and respond to patient demand. Franco and Alfonso-Lizarazo [25] illustrated a simulation–optimization procedure based on the stochastic counterpart that is taken into account for optimizing PSCN. This model proposed the maximum possible date of expiration and minimized the entire amount of expired drug. Goodarzian et al. [26] described a novel multi-objective, multi-period PSCN model in an uncertain condition. They suggested various metaheuristic algorithms, including Multi-Objective Social Engineering Optimization (MOSEO), Multi-Objective Simulated Annealing (MOSA), Multi-Objective Keshtel Algorithm (MOKA), and Multi-Objective Firefly Algorithm (MOFFA). Kadkhodamanesh et al. [27] assessed Iran's GDP share of pharmaceuticals based on the factors that influence expenditure and compared it to that of 31 other Organization for Economic Cooperation and Development (OECD) members. Priyan and Mala [28] suggested a game theory procedure for employing inventory policy for overseeing the flow of RM combining multiple quality features in PSC. Rekabi et al. [2] devised a multi-objective model with the goal of reducing overall expenses spent in the SC network and the total elapsed time (responsiveness), along with the detrimental impacts of $CO_2$ emissions, and taking congestion in production centers and work scheduling in flexible flow shop systems into consideration. Goodarzian et al. [3] suggested a novel multi-objective optimization procedure for the PSCND problem that is intended to maximize the dependability of the transportation system and diminish the overall cost and delivery time of pharmaceutical commodities to pharmacies.

Konovalenko and Ludwig [29] used real data from a global logistics service provider over a five-year period to assess the best performance of ten ML classification techniques in a case study of temperature warnings in the PSC. Ahmadi et al. [30] developed intelligent inventory management procedures for managing perishable pharmaceutical items in a healthcare system comprising a number of local hospitals and warehouses. Sazvar et al. [31] developed a scenario-based programming model to design sustainable closed-loop PSCs, which examines three different groups of reverse flows of expired medications. Iacocca et al. [32] suggested the potential for leveraging coordinated, medication-specific relationships between independent chain and mail-order pharmacies to lower prescription costs and/or boost SC profit.

Mediavilla et al. [33] described an evaluation of the AI techniques published in the previous five years (2017–2021). Additionally, a classification that defines the trend of the used AI methods is illustrated by the clustering technique. Finally, a classification of the various AI techniques based on the volume, dimensionality, and time horizon of the forecast is given. Tucker and Daskin [34] presented the first model of PSC reliability with three key procedures that firms may use to enhance reliability, and they parametrized the model using data from significant medication shortage databases and a case example of a generic oncology medication injection. Saha et al. [35] offered a framework on how to include and support Pharma 4.0 to gain sustainability in SCs, as well as experimental research on the relation between new technologies and PSC performance. Malleeswaran and Uthayakumar [36] highlighted the need for two-tier PSC management in the healthcare sector along with visiting frequency, level of service, and household energy usage.

Ahmad [37] considered sustainable goals in the PSC optimization structure with several constraints under uncertainty. The trade-off between socio-economic and environmental goals is declared by ensuring the best allocation of several products among various echelons. According to the literature summary, the PSCND problem has been investigated in a number of studies in recent years from various perspectives. Nonetheless, there is no comprehensive research work investigating the PSCND problem by considering STW and perishability by applying ML technique for predicting demand at the same time. Feng et al. [38] introduced a particular structure to develop pharmaceutical

**Table 1**
Literature summary.

| Author | Objective | | Objective function | | Type of problem | | | | Nature of model | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Single | Multi | Cost | Responsiveness | Optimization | Distribution | Inventory | Location/Allocation | Deterministic | Uncertainty | TMs | Perishability | ML | Numerical example | Solution approach |
| Savadkoohi et al. (2018) | ✘ | ✔ | ✔ | ✘ | ✘ | ✘ | ✘ | ✘ | ✔ | ✘ | ✘ | ✔ | ✘ | ✘ | Exact Method |
| Jabbarzadeh et al. (2019) | ✘ | ✔ | ✔ | ✘ | ✔ | ✔ | ✔ | ✘ | ✘ | ✔ | ✔ | ✘ | ✘ | ✘ | Cplex |
| Franco et al. (2020) | ✘ | ✔ | ✘ | ✘ | ✔ | ✘ | ✔ | ✘ | ✘ | ✔ | ✘ | ✘ | ✘ | ✔ | Epsilon-Constraint |
| Goodarzian et al. (2020) | ✘ | ✘ | ✔ | ✘ | ✔ | ✘ | ✔ | ✔ | ✔ | ✘ | ✘ | ✘ | ✘ | ✔ | Hybrid Metaheuristic Algorithms |
| Rekabi et al. (2021) | ✘ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✘ | ✘ | ✔ | ✘ | ✔ | LP-metric & GA |
| Goodarzian et al. (2021) | ✘ | ✘ | ✔ | ✘ | ✔ | ✔ | ✔ | ✔ | ✘ | ✔ | ✘ | ✘ | ✘ | ✔ | Hybrid metaheuristic algorithms |
| Ahmadi et al. (2022) | ✔ | ✘ | ✔ | ✘ | ✔ | ✘ | ✔ | ✘ | ✘ | ✔ | ✘ | ✔ | ✔ | ✘ | Q-learning& Reinforcement learning |
| Sazvar et al. (2022) | ✘ | ✔ | ✘ | ✘ | ✔ | ✔ | ✔ | ✔ | ✔ | ✘ | ✘ | ✔ | ✘ | ✔ | Hybrid method |
| Tucker & Daskin, (2022) | ✔ | ✘ | ✘ | ✘ | ✔ | ✘ | ✘ | ✘ | ✘ | ✔ | ✘ | ✘ | ✘ | ✔ | Cplex |
| Kumar et al. (2023) | ✔ | ✘ | ✘ | ✘ | ✔ | ✔ | ✘ | ✘ | ✘ | ✘ | ✘ | ✘ | ✔ | ✘ | ML |
| Abdallah and Nizamuddin (2023) | ✘ | ✘ | ✘ | ✘ | ✔ | ✔ | ✘ | ✘ | ✘ | ✘ | ✘ | ✘ | ✘ | ✘ | Block chain and Internet of Thing |
| This study | ✘ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✔ | ✘ | ✔ | ✔ | ✔ | ✔ | Goal Attainment |

Technology Roadmaps (TRMs). This TRM suggests three improvements over the standardized TRM. Designing the layers is the initial extension. The choice of the data source is the second extension. And the final improved version is the data source processing. Kumar et al. [39] developed a vendor incoterm (contract) selection model for ML for direct drop-shipping in a worldwide omnichannel PSC. Abdallah and Nizamuddin [40] proposed a decentralized blockchain-based structure with no intermediaries like pharmacies or hospitals for the online sale of pharmaceutical products. Daryanian et al. [41] introduced a novel fuzzy robust stochastic model to deal with the crisis logistics of medicinal products by using the sustainability and resilience dimensions. Kochakkashani et al. [42] provided a mathematical model to support PSC planning during the COVID-19 epidemic. Also, they offered new insights into hospital supply chain problems by unifying cold and non-cold chains and considering a wide range of pharmaceuticals and vaccines. Yang [43] presented an environmental, social, and governance strategic model to manage PSCs with financial obstacles. Lindahl et al. [44] proposed a method for setting up and solving integrated capacity and production planning problems for a given manufacturing network. Shakouhi et al. [45] provided a competitive PSC under a product life cycle and offered Nash and Stackelberg games under the marketing mix strategies and product life cycle.

The literature summary that is presented in Table 1 highlights that there are various works in the literature that have studied the PSCND problem, but only a few of them have used perishability. This study extends the PSCND model by considering perishability and STW policy. Moreover, ML algorithms are used and compared to predict demand. Then GA procedure is utilized to solve the problem.

### 2.1. Contribution of this work

Based on Table 1, the literature summary demonstrates that the literature contains a number of studies that studied the PSCND problem, but there is no research that has considered perishability with STW mechanism for minimizing the penalty due to the delivery schedule violation. According to the author's knowledge, only a few studies projected demand to be used to the model of the study, despite the fact that this parameter is crucial, and there is no paper that uses ML techniques (LR and QR) in the mathematical model in PSC to make it more realistic. In this work, the estimated demand enters the mathematical modeling as a parameter. Then the model is solved using GA approach and some numerical examples, and several sensitivity analyses on important parameters are implemented to validate the suggested model. This work aims to develop a bi-objective model to reduce costs, reduce delivery times in DCs, reduce $CO_2$ emissions by considering different TMs, and decrease the quantity of unsatisfied demands and wastes to design an efficient and responsive PSCN that were not considered in past.

This study extends the PSCN model by considering perishability, STW, different Transportation Modes (TMs) for shipping products, and forecasting demand with the QR algorithm with RStudio software.

Overall, the major contributions of this work are as follows:

- Designing a novel multi-objective, multi-commodity, multi-echelon PSCND problem in which both strategic and tactical decisions are made by forecasting demand using a QR technique and considering perishability and STW policy for the first time.
- Decreasing the quantity of unsatisfied demands and waste in the model.
- Considering the shelf-life of products because medicine is a perishable product.
- Proposing the LR and QR techniques to predict demand and comparing them. Finally, selecting method with better performance in this PSCN model to increase the accuracy of the model.
- The GA algorithm has been utilized to solve the suggested model.
- Finally, by solving the proposed model, and validating, we find the best model that can be used in pharma industry.

## 3. Demand forecasting

In this section the forecasting techniques is described.

### 3.1. Data

The data includes medicine demand for between 2013 and 2022. We want to predict demand for 2023 in this work. So, the two forecasting techniques is considered. Additionally, the empirical data provided for this paper gathered from some pharma center in Tehran and experts. Fig. 1 shows the flow diagram of steps implemented in the ML models.

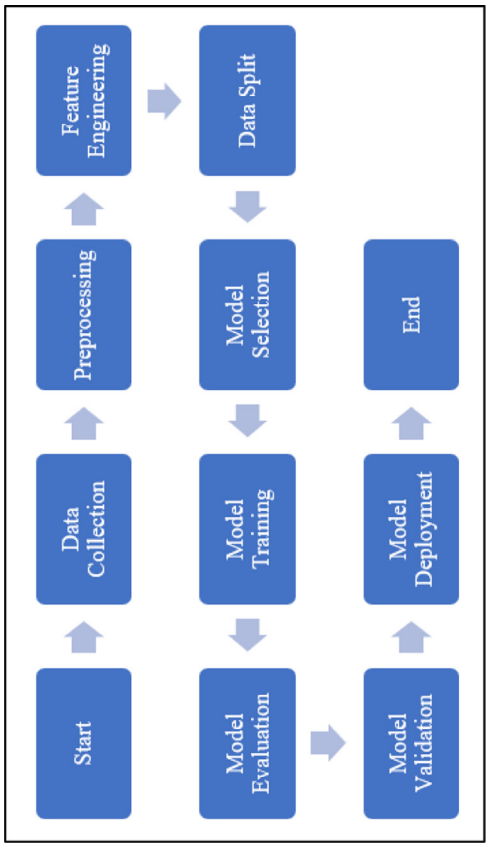


**Fig. 1.** The flow diagram of steps implemented in the ML model.

### 3.2. Linear regression

The LR technique for predicting discreetly and clearly displays connections through two variables by utilizing a linear equation to fit training data. LR indicates a linear relation through the exploratory variable and the dependent variable. If just one input variable is examined, this format of LR is referred to as "basic LR". The best-fit line LR is computed within a common slope intercept [46].

Where $y = The\ Dependent\ Variable$, $x = The\ Independent\ Variable$, $a_0 = line\ interception$, $a_1 = LR\ coefficient$ and $" = error$. The structure of the formula following:

$$y = a_0 + a_1 x + "$$

For forming the LR algorithm, first, the data set has to be imported into the RStudio software. After that, the preparation step has been done. Then the data were separated into two groups named "train" and "test". Finally, the LR model was built on train data, and pharmaceutical product demand is forecasted. The model is as follows:

$$Model1 < -lm(Medicine{:}demanded \sim AQI; data = train)$$

We have utilized LR and forecasted demand with the Air Quality Index (AQI).

### 3.3. Quadratic regression

QR is an augmentation of uncomplicated LR that is utilized to discover the uniform line's equation that best fits a group of data. The decision of specifying which parabola equation best fits a group of data is called QR [47,48]. At the end, we obtain an equation of the form:

$$y = a_0 + a_1 x + a_2 x^2 + "$$

$$Model2 < -lm(Medicine{:}demanded \sim AQI + I(AQI, 2); data = train)$$

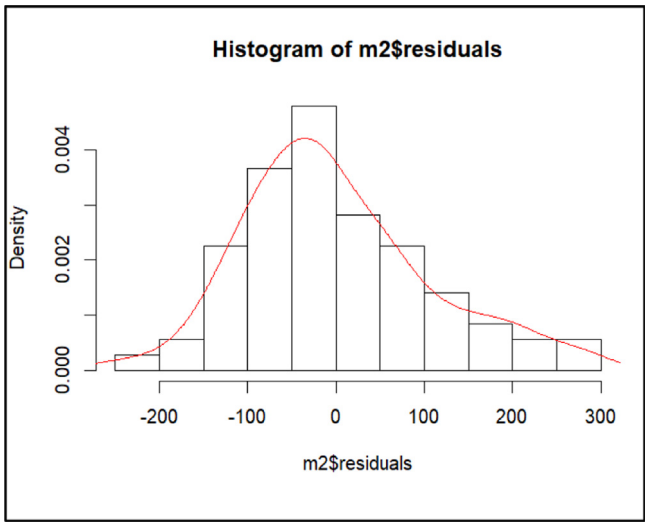


**Fig. 2.** Residual distributed.

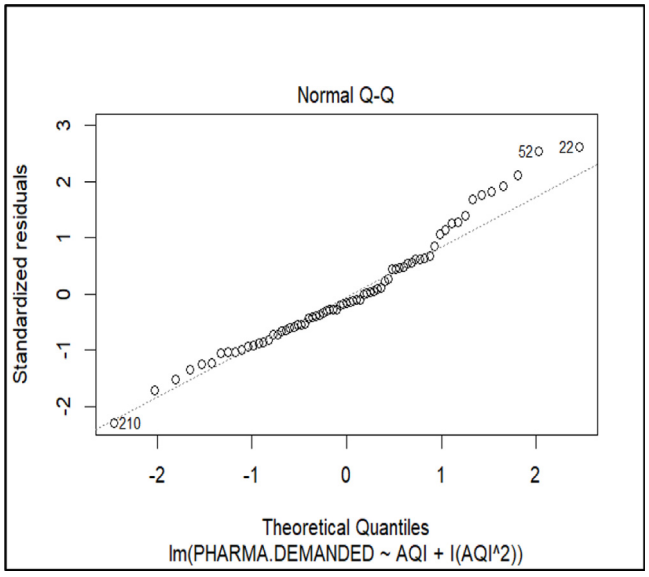


**Fig. 3.** Residual distributed.

The coefficient that results from QR and residual distributions is as follows (Figs. 2 and 3): $a_0 = 630{:}066$, $a_1 = 4{:}89$, and $a_2 = -0.008$. Fig. 2 illustrates the residual distribution of the QR model after running the model has not shown normal distribution exactly but is much close to being normal so it has a good condition. Fig. 3 shows residual scattered near desired line and just a few data in the right side are not near line. This condition demonstrates the good performance of the model. You can see if the residuals are reasonably close to normal via a Q-Q plot. Fig. 2 (Q-Q plot) shows residual scattered near desired line and just a few data in the right side are not near line. This condition demonstrates the good performance of the model. As we build our ML model, ensure we check the distribution of the error terms or prediction error using a Q-Q plot. If there is a significant deviation from the mean, we might want to check the distribution of our feature variable and consider transforming them into a normal shape. The LR model does not have good performance because its p-value is less than 0.05, which rejects the normality assumption. So, we have considered QR, which has better performance than LR, by comparing them in Table 2.

According to Table 2, QR has a bigger Adjusted R-squared, so it is better than LR, and its p-value is larger than 0.05. By comparing

**Table 2**
Comparison two ML model.

| ML model | R-squared | p-value |
|---|---|---|
| LR | 0.2334 | 0.0016 |
| QR | 0.6668 | 0.4573 |

these two methods, the QR method is better than the LR. So, the QR technique is utilized to predict the demand for our study.

### *3.4. What merits does QR have over a Principal Component Analysis (PCA) or even a stepwise regression model?*

Regression examines the relationship between a response variable and a number of explanatory variables, whereas PCA looks for relationships within a collection of variables. To enable the model to be used to describe or forecast particular outcomes, the relationship between two sets of variables is modeled utilizing QR. In contrast to PCA, which merely accepts data and ignores any dependent variables, QR is, generally speaking, a supervised learning technique. In the case of a dependent variable, a supervised approach would be appropriate for your objectives. Another disadvantage of PCA is that it is also sensitive to outliers. Regression models are also simple to comprehend because they are based on fundamental statistical concepts like correlation and least-square error. Regression models also produce an algebraic equation that can be used to predict with ease.

## 4. Problem description and mathematical formulation

This section starts with a description of the issue and then goes about great detail into the mathematical model.

### *4.1. Problem description*

This work aims to structure a multi-product, multi-commodity, lean-responsive PSCND considering forecasting techniques. The suggested PSCN include four levels consisting of suppliers, Manufacturing Sites (MSs), DCs and Customer. Beginning the material flow process is the acquisition of the required Raw Materials (RMs) from vendors. The RMs are then assembled in the MSs to create the medicines, which are then sent to demand points by DCs. There are different types of TMs to deliver products. To design the above-mentioned PSCN, a BOMILP model is developed. Afterwards, the GA procedure is proposed to solve it. The first Objective Function (OF) seeks to reduce overall expenditures such as facility establishment costs, operational costs, transportation costs, and holding costs. The second OF considers aims at minimizing the punishment for violating the delivery schedule. To construct the STW system, we let $E_{pk}$ and $L_{pk}$ denote the Earliest Delivery Dates (EDDs) and Latest Delivery Date (LDDs) respectively, for product p delivered by DC k. Generally, components will be received if delivered within $E_{pk}$ and $L_{pk}$. To relax the limited TW, $LE_{pk}$ is set to denote the lower bound of the EDD and $UL_{pk}$ denotes the upper bound of the LDD. All components that have been delivered before $LE_{pk}$ will be refused and returned to the DCs. Although the components delivered in the TW of $LE_{pk}$ and $E_{pk}$ as well as the TW of $L_{pk}$ and $UL_{pk}$ will be received, a specific penalty will be carried to DC k. In this work, the penalty function for product p delivered by DC k (i.e., $PE_{pk}$) uses the concept from Ioannou et al. [49] namely

$$PE_{pk} = \begin{cases} \infty & if\ lt_{pk} < LE_{pk} \\ cle_{pk}\ E_{pk} - lt_{pk} & if\ LE_{pk} \le lt_{pk} \le E_{pk} \\ 0 & if\ E_{pk} \le lt_{pk} \le L_{pk} \\ cul_{pk}\ lt_{pk} - L_{pk} & if\ L_{pk} \le lt_{pk} \le UL_{pk} \\ \infty & if\ UL_{pk} \le lt_{pk} \end{cases}$$

Additionally, ML method have used to predict the demand of customers first time in this study. By predicting the demand model become more responsive and shortage in PSCN is decreased. In order to formulate the model, the following assumptions are considered. In addition, indices, parameters, and decision variables are illustrated after assumptions.

### *4.2. Assumption*

The following assumptions are utilized to construct the model:

- All medicines cannot be produced by each MS.
- Suppliers', DCs', and customers' locations and numbers are not given. These are variables that are obtained after solving the model.
- Each medication has an expiration date that must be met.
- The shelf-life of each product must be longer than its lead time to decrease shortage and waste.
- Several TMs are available and can be used according to the demand.
- The facilities of the SC (i.e., DCs) have finite capacities.
- A certain amount of waste can be produced in each period by specifying the maximum amount of waste for each DC.
- Considering the ML method to predict the demand of customers.

*Indices:*

| | |
|---|---|
| p | Index of medicines |
| m | Index of RMs |
| s | Index of suppliers |
| *j* | Index of MSs |
| k | Index of DCs |
| c | Index of customers |
| t | Index of Time Period (TP) |
| r | Index of Production Period (PP) |
| e | Index of Delivery Period (DP) |
| h | Index of TMs |

*parameters:*

| | |
|---|---|
| $ec_j$ | Establishment cost of MS j |
| $eck_k$ | Establishment cost of DC k |
| $ho_{pj}$ | Holding cost of medicine p at the MS j |
| $hok_{pk}$ | Storing cost of medicine p in DC k |
| $tc_{sj}^{mh}$ | Cost of transferring RM m from supplier s to MS j by TM h |
| $tck_{jk}^{ph}$ | Cost of transferring medicine p from MS j to DC k by TM h |
| $tcc_{kc}^{ph}$ | Cost of transferring medicine p from DC k to customer c via TM h |
| $d_{cpt}$ | The overall demand of customer c for medicine p, at TP t |
| $sst_{kpt}$ | Safety stock of the medicine p, at TP t in the DC k |
| $us_p$ | The lifespan of medicine p |
| $mcap_{jp}$ | The maximum capacity of MS j for medicine p |
| $mcapp_k$ | The maximum capacity of DC k |
| $sc_p$ | Shortage cost for a unit of medicine p |
| $di_{sj}$ | The space between supplier s and MS j |
| $dj_{jk}$ | The space between MS j and the DC k |
| $dk_{kc}$ | The space between DC k and customer c |
| $cr_{mp}$ | Consumption rate of RM m in unit medicine p |
| $mc_{pj}$ | Production cost of MS j for medicine p |
| $mqw_{kt}$ | Maximum waste in DC k at TP t |
| $E_{pk}$ | EDD for medicine p from DC k |
| $L_{pk}$ | LDD for medicine p from DC k |
| $lt_{pk}$ | lead time for product p from DC k |

| | |
|---|---|
| $LE_{pk}$ | Lower bound for the EDD for medicine p from DC k. No delivery is allowed prior |
| $UL_{pk}$ | Upper bound for the LDD for medicine p from DC k. No delivery is allowed after |
| $cle_{pk}$ | Unit penalty cost initiated if medicine p is delivered between $LE_{pk}$ and $E_{pk}$ by DC k |
| $cul_{pk}$ | Unit penalty cost initiated if medicine p is delivered between $L_{pk}$ and $UL_{pk}$ by DC k |

*Variables:*

| | |
|---|---|
| $xp_j$ | 1 if a MS is established in location j otherwise zero |
| $yd_k$ | 1 if a DC is established in location k, otherwise zero |
| $il_{jptr}$ | Inventory level of medicine p in MS j at TP t produced at PP r |
| $ill^{kp}_{tre}$ | Inventory level of medicine p at DC k at TP t produced at PP r received at DP e |
| $fr_{sjmht}$ | The flow of RM m between supplier s and MS j by TM h at TP t |
| $fo_{hjkpt}$ | The flow of medicine p between MS j and DC k at TP t by TM h |
| $uc_{kcpth}$ | The flow of medicine p between DC k and customer c at TP t via TM h |
| $qud_{cpt}$ | Quantity of unsatisfied demands from demand zone c related to medicine p in TP t |
| $qw_{kpt}$ | Quantity of waste from DC k related to medicine p in TP t |

### 4.3. Mathematical modeling

$$\begin{aligned}\min z1 = {} & \sum_{j,l} ec_j \times xp_{jl} + \sum_{k} eck_k \times yd_k + \sum_{j}\sum_{p}\sum_{t}\sum_{r=1}^{t} ho_{pj} \times il_{jptr} \\ & + \sum_{j}\sum_{k}\sum_{p}\sum_{t}\sum_{h} tck^{ph}_{jk} \times dj_{jk} \times fo_{hjkpt} \\ & + \sum_{k}\sum_{p}\sum_{t}\sum_{e=r}^{t}\sum_{r=1}^{t} hok_{pk} \times ill^{kp}_{tre} \\ & + \sum_{s}\sum_{j}\sum_{p}\sum_{h}\sum_{m}\sum_{t} tc^{mh}_{sj} \times di_{sj} \times fr_{sjmht} \\ & + \sum_{k}\sum_{h}\sum_{p}\sum_{c}\sum_{t} tcc^{ph}_{kc} \times dk_{kc} \times uc_{kcpth} + \sum_{j}\sum_{p} mc_{pj} \times xp_j \\ & + \sum_{k}\sum_{p}\sum_{t} sc_p \times qud_{cpt} \end{aligned} \quad (1)$$

$$\begin{aligned}\min z2 = {} & \sum_{k}\sum_{c}\sum_{p}\sum_{t}\sum_{h} uc_{kcpth} cle_{pk} \quad E_{pk} - lt_{pk} \\ & + \sum_{k}\sum_{c}\sum_{p}\sum_{t}\sum_{h} uc_{kcpth} cul_{pk} \quad lt_{pk} - L_{pk} \end{aligned} \quad (2)$$

The first Eq. (1) diminished the overall costs of PSCN. Foundation costs, holding costs, distribution and production costs, transportation and shortage costs are all included in the PSC network. The second Eq. (2) minimizes the penalty depending on the delivery schedule violation.

*Constraint:*

$$\sum_{r=1}^{t} il_{jptr} = \sum_{h}\sum_{s} fr_{sjmht} - \sum_{k}\sum_{h}\sum_{l} fo_{hjklpt} \quad (3)$$
$$\forall j; p; t = 1 < us_p$$

$$\sum_{r=1}^{t} il_{jptr} = \sum_{r=1}^{t-1} il_{jpt-1r} + \sum_{h}\sum_{s} fr_{sjmht} - \sum_{k;h} fo_{hjkpt} \quad (4)$$
$$\forall j; p; 1 < t < us_p$$

$$\sum_{r=t+1-us_p}^{t} il_{jptr} = \sum_{r=t+1-us_p}^{t-1} il_{jpt-1r} + \sum_{h}\sum_{s} fr_{sjmht} - \sum_{k}\sum_{h}\sum_{l} fo_{hjkpt} \quad (5)$$
$$\forall j; p; t \geq us_p$$

$$\sum_{e=r}^{t}\sum_{r=1}^{t} ill^{kp}_{tre} = \sum_{j}\sum_{h} fo_{hjkpt} - \sum_{c}\sum_{h} uc_{kcpth} \quad (6)$$
$$\forall k; p; t = 1 < us_p$$

$$\sum_{e=r}^{t}\sum_{r=1}^{t} ill^{kp}_{tre} = \sum_{e=r}^{t-1}\sum_{r=1}^{t-1} ill^{kp}_{t-1re} + \sum_{j}\sum_{h} fo_{hjkpt} - \sum_{c}\sum_{h} uc_{kcpth} \quad (7)$$
$$\forall k; p; 1 < t < us_p$$

$$\begin{aligned}\sum_{e=r}^{r+us_p-1}\sum_{r=t-us_p+1}^{t} ill^{kp}_{tre} = {} & \sum_{e=r}^{t-1}\sum_{r=t-us_p+1}^{t-1} ill^{kp}_{t-1re} + \sum_{j}\sum_{h} fo_{hjkpt} \\ & - \sum_{c}\sum_{h} uc_{kcpth} \end{aligned} \quad (8)$$
$$\forall k; p; t \geq us_p$$

$$\sum_{h}\sum_{j} fo_{hjkpt} - \sum_{c} d_{cpt} - \sum_{r}\sum_{e} ill^{kp}_{tre} + \sum_{c} qud_{cpt} = qw_{kpt} \quad (9)$$
$$\forall k; p; t \geq us_p$$

$$\sum_{p} qw_{kpt} \leq mqw_{kt} * yd_k \quad \forall k; t \quad (10)$$

$$\sum_{j}\sum_{h}\sum_{t}\sum_{p} fo_{hjkpt} \leq mcap p_k * yd_k \quad \forall k \quad (11)$$

$$\sum_{r}\sum_{t}\sum_{e} ill^{kp}_{tre} \geq \sum_{t} sst_{kpt} * yd_k \quad \forall k; p \quad (12)$$

$$\sum_{h}\sum_{s} fr_{sjmht} = \sum_{k;h;p} cr_{mp} \times fo_{hjkpt} \quad \forall t; m; j \quad (13)$$

$$\sum_{k;k';h} uc_{kcpth} \geq d_{cpt} \quad \forall c; p; t \quad (14)$$

$$\sum_{h;j} fo_{hjkpt} \geq \sum_{c} uc_{kcpth} \quad \forall k; p; t \quad (15)$$

$$il_{jptr} = 0 \quad \forall j; p; t; r; t < r \quad (16)$$

$$ill^{kp}_{tre} = 0 \quad \forall k; p; t; r; e; t < r \quad (17)$$

$$ill^{kp}_{tre} = 0 \quad \forall k; p; t; r; e; e < r \quad (18)$$

$$us_p \geq lt_{pk} \quad \forall p; k \quad (19)$$

$$xp_j, yd_k \in \{0; 1\} \quad (20)$$

$$il_{jptr}; ill^{kp}_{tre}; fr_{sjmht}, fo_{hjkpt}, uc_{kcpth} \geq 0 \quad \forall c; p; j; m; s; k; k'; t; h \quad (21)$$

Constraints (3)–(5) indicate the MS's inventory level at various points in the product's lifetime. Since we are at the beginning of the TP in Constraint (3), the inventory of the previous stages is zero. However, in Constraints (4) and (5), there is already some inventory remaining. The inventory level at DCs is shown in Constraints (6)–(8) at various time intervals. The product's lifespan is the only difference between these three constraints; in fact, the r (PP) and e (DP) intervals are determined by the product's lifetime. We can say that a medicine is delivered when it has been produced during the previous TP because of these intervals for delivery and production. The DP and PP intervals will include the product's lifespan if the TP is longer than the product's lifespan. Constraint (9) calculates the amount of waste produced in each TP. Constraint (10) demonstrates the allowable amount of waste that can be produced in each DC during a TP. Constraint (11) demonstrates the capacity of DCs and asserts that pharmaceutical products

can be moved to DCs where DCs have been established. Constraint (12) indicates that DCs should always have a minimum safety stock that can respond to customer needs. The number of raw materials purchased from suppliers is calculated using constraint (13). According to constraint (14), the amount of product transferred from the DC to the customer must not be less than the required demand. Constraint (15) illustrates that the number of products transferred from the MS to the DC cannot be less than the related amounts transferred to the customer. Inventory is at a zero level in constraints (16)–(18). Constraint (19) denotes shelf-life of product must become bigger than lead time. Constraints (20) and (21) are related to binary variables, and positive continuous variables, respectively.

## 5. Solution approach

This approach seeks to detect a solution that minimizes the maximum deviation (Z) between each OF and overall values. The GA procedure is a multi-objective problem-solving method that takes action by getting initial information from the Decision Maker (DM). In the GA approach, the DM first sets the target level ($g_j$) for each OF and then allocates the number of weights ($w_j$) to each goal. Note that the weights $w_j$ are specified as $\sum_j w_j = 1$. The GA model is structured as follows:

$$\begin{aligned}
&\min z_1(x) \\
&\min z_2(x) \\
&\quad \cdots \\
&\min z_k(x) \\
&\text{s.t.:} \\
&x \in \Omega
\end{aligned}
\qquad
\begin{aligned}
&\\
&\\
&\\
&\\
&\min Z \\
&\text{s.t.:} \\
&z_j(x) - w_j \times Z \le g_j \quad \forall j = 1, \ldots, k \\
&x \in \Omega
\end{aligned}$$

The GA model has been utilized in this study because it has a number of benefits.

- Simplicity and quick calculation times.
- It can generate a variety of ideal Pareto solutions by adjusting the weights.

The reader is directed to these sources to learn more about this approach [50,51].

## 6. Numerical results

The BOMILP model is coded in GAMS 27.2 commercial software and solved by a CPLEX solver. GA method is executed on a computer with Intel (R) Core (TM) i7-10750H CPU @ 2.60 GHz. All parameters are randomly generated in Excel and denoted in Table 3. Different sizes of problems are shown in Table 4, and numerical results are denoted in Table 5.

This section focuses on solving some examples of the problems suggested in the previous section and validating the proposed model. To do so, the numerical findings obtained by the GA approach are analyzed concerning the primary measures: the first OF value ($Z_1$), the second OF value ($Z_2$), and CPU-time. The numerical findings are displayed in Tables 5 and 6. In Table 5, the OF's weights are ($w_1 = 0.5; w_2 = 0.5$).

According to Table 6, regarding the GA approach, when the weights of the OF (w) grow, much Z is observed since when the weights become large, the model aims to maximize significant parts, which are the first and second OF. So, the model is valid.

**Table 3**
Parameters in the test problem.

| Parameters | Value | Parameters | Value |
|---|---|---|---|
| $ec_j$ | ~uniform (2000,3000) $ | $di_{sj}$ | ~uniform (30,100) km |
| $eck_k$ | ~uniform (2000,3000) $ | $dj_{jk}$ | ~uniform (30,100) km |
| $ho_{pj}$ | ~uniform (300,2500) $ | $dk_{kc}$ | ~uniform (100,5000) km |
| $hok_{pk}$ | ~uniform (300,1000) $ | $cr_{mp}$ | ~uniform (2,4) |
| $tc^{mh}_{sj}$ | ~uniform (200,500) $ | $mc_{pjl}$ | ~uniform (30,100) $ |
| $tck^{ph}_{jk}$ | ~uniform (200,500) $ | $E_{pk}$ | ~uniform (200,500) min |
| $tcc^{ph}_{kc}$ | ~uniform (50,1500) $ | $L_{pk}$ | ~uniform (200,500) min |
| $d_{cpt}$ | ~uniform (100,3000) kg | $lt_{pk}$ | ~uniform (200,500) min |
| $sst_{kpt}$ | ~uniform (10,100) kg | $LE_{pk}$ | ~uniform (200,500) min |
| $us_p$ | ~uniform (6,12) month | $UL_{pk}$ | ~uniform (200,500) min |
| $mcap_{jp}$ | ~uniform (100,2000) kg | $cle_{pk}$ | ~uniform (200,500) $ |
| $mcapp_k$ | ~uniform (100,2000) kg | $cul_{pk}$ | ~uniform (200,500) $ |

**Table 4**
Test problems.

| Index | Description | Problem | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| p | No. of products | 3 | 3 | 4 | 5 | 5 |
| m | No. of RMs | 2 | 2 | 2 | 3 | 3 |
| s | No. of supplier | 4 | 5 | 6 | 6 | 6 |
| j | No. of MS | 3 | 4 | 5 | 6 | 7 |
| k | No. of DC | 3 | 4 | 5 | 6 | 8 |
| c | No. of customer | 3 | 4 | 6 | 8 | 10 |
| t | No. of TP | 3 | 3 | 5 | 5 | 6 |
| r | No. of PP | 2 | 2 | 2 | 3 | 3 |
| e | No. of DP | 2 | 2 | 2 | 3 | 3 |
| h | No. of TM | 2 | 2 | 2 | 3 | 3 |

**Table 5**
Computational finding obtained for each test.

| Test problem | OF | | | CPU–time |
|---|---|---|---|---|
| | $Z_1$ | $Z_2$ | $Z_{GA}$ | |
| 1 | 3.020242E+9 | 5 915 555.204 | 1 239 356.409 | 3 |
| 2 | 4.112853E+9 | 1.141134E+7 | 2 719 143.159 | 6 |
| 3 | 2.19444E+10 | 1.228052E+7 | 3 774 021.904 | 10 |
| 4 | 6.74635E+10 | 1.595851E+7 | 6 917 080.367 | 79 |
| 5 | 1.05756E+11 | 2.457855E+7 | 7 276 602.407 | 1268 |

**Table 6**
Computational finding obtained for problem No. 3.

| | Weights | | Objective function | |
|---|---|---|---|---|
| | $w_1$ | $w_2$ | $Z_1$ | $Z_2$ |
| 1 | 0.5 | 0.5 | 2.19444E+10 | 1.228052E+7 |
| 2 | 0.4 | 0.6 | 2.19439E+10 | 1.246806E+7 |
| 3 | 0.3 | 0.7 | 2.19435E+10 | 1.262658E+7 |
| 4 | 0.2 | 0.8 | 2.19431E+10 | 1.276233E+7 |
| 5 | 0.1 | 0.9 | 2.19428E+10 | 1.287990E+7 |

### 6.1. Sensitivity analysis

*Analysis based on maximum capacity for MS ($mcap_{jp}$)*

As the capacity increases, both the first and second OFs diminish. That makes sense because controlling the Supply Chain Network (SCN) becomes simpler as capacity rises (Fig. 4). It means that when the capacity of MSs increases, the SCN needs fewer MSs. So, the total cost decreased. Furthermore, the inventory level will increase, which makes the chain responsive.

*Analysis based on shelf-life of medicine ($us_p$)*

As shown in Fig. 5, a rise in shelf life decreases the total costs and penalties depending on the delivery schedule violation. By increasing the shelf life of medicines, the shortage and delivery time decrease. So, the total cost decreases, and customer demand will be met at the right time. It means that when the shelf life of medications is increased, they